



**Killara**  
HIGH SCHOOL

STUDENT NUMBER

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# MATHEMATICS EXTENSION 2

ASSESSMENT TASK 1

Monday 6th December, 2010

**TIME ALLOWED:** *Reading time: 5 minutes*  
*Working time: 60 minutes*

**INSTRUCTIONS:** This paper consists of 2 questions  
ANSWER all two questions  
Start each question in a new booklet  
Marks may not be given for untidy and poorly arranged work.  
All relevant working should be shown.

***THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM***

**OUTCOMES ASSESSED:**

|      |                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------|
| ✓ E1 | Appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems              |
| ✓ E3 | Uses the relationship between algebraic and geometric representations of complex numbers and of conic sections. |
| ✓ E9 | Communicates abstract ideas and relationships using appropriate notation and logical argument.                  |

|                            | MARKS |
|----------------------------|-------|
| QUESTION 1 Complex Numbers | /15   |
| QUESTION 2 Complex Numbers | /17   |
|                            | /32   |

***This assessment task constitutes 10% of the Higher School Certificate Course Assessment.***

**Total marks - 32**

**Attempt questions 1 – 2**

Answer each question on a separate writing booklet.

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**Question 1** (15 marks) Use a SEPARATE writing booklet

**Marks**

(a) Given  $z_1 = 3 - 2i$ ,  $z_2 = 1 + 3i$ . Find the value of

(i)  $z_1 + z_2$  **1**

(ii)  $z_1^2$  **1**

(iii)  $\frac{z_1}{z_2}$ . Leave your answer in Cartesian form. **2**

(b)  $z = 1 - i$ .

(i) Express  $z$  in modulus-argument form. **2**

(ii) Hence, determine the smallest value of  $n$  for which  $z^n$  is a purely imaginary number, where  $n$  is a positive integer. **2**

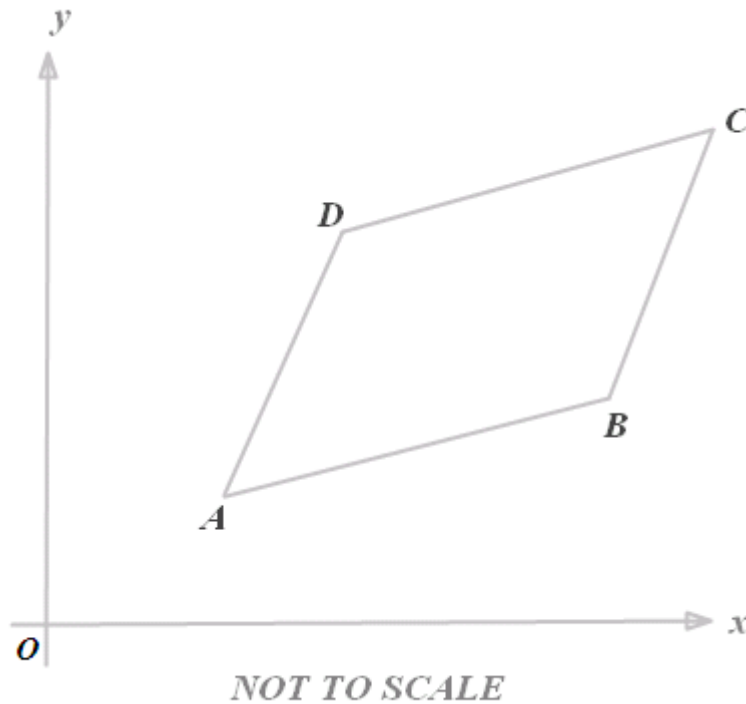
(c) Find the square root of  $-4 + 3i$  **3**

(d) Prove that for all complex numbers  $z$  and  $w$ , **4**

$$|z + w|^2 + |z - w|^2 = 2(|z|^2 + |w|^2).$$

Give a geometrical interpretation of this equation in the complex plane.  
(Use a diagram to aid your explanation)

- (a)  $A$  and  $B$  are points in the Argand diagram representing the numbers  $z_1 = 3 + i$ ,  $z_2 = 3 + 2\sqrt{3} + 3i$  respectively.



- (i) Find the complex number that represents the vector  $\overrightarrow{AB}$ . 1
- (ii)  $ABCD$  is a parallelogram where  $AB = 2AD$  and  $\angle DAB = \frac{\pi}{6}$ . 3  
Find the complex number that represents  $D$ .

**Question 2 Continued.....****Marks**

(b)  $\text{Arg}(z - 2) = 2\text{Arg}(z)$

(i) Show vectors representing  $z$ ,  $z - 2$  in an Argand diagram. **1**

(ii) If the point  $P$  represents  $z$ ,  $O$  is the origin and  $A$  has coordinates  $(2, 0)$  **1**  
in this Argand diagram, what is the nature of  $\triangle OAP$  for non-real  $z$ ?

(iii) Deduce that if  $z$  is non-real, then  $P$  lies on a circle and state its centre and radius. **2**

On a new diagram, sketch the locus in the Argand diagram of a point representing  $z$  satisfying  $\text{Arg}(z - 2) = 2\text{Arg}(z)$ , for both real and non-real  $z$ .

(c) (i) Use De Moivre's theorem to show that  $(\cot \theta + i)^n + (\cot \theta - i)^n = \frac{2 \cos n\theta}{\sin^n \theta}$ . **1**

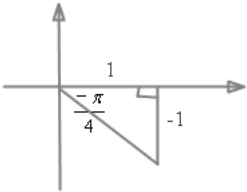
(ii) Show that the equation  $(x + i)^5 + (x - i)^5 = 0$  has the roots  $0, \pm \cot \frac{\pi}{10}, \pm \cot \frac{3\pi}{10}$ . **3**

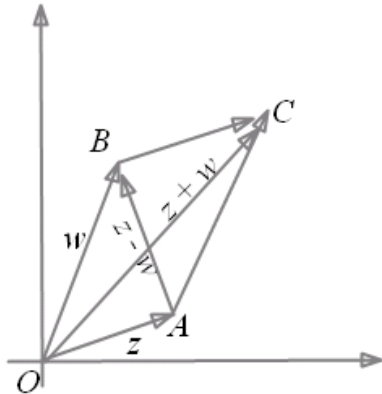
(iii) Hence, show that the equation  $x^4 - 10x^2 + 5 = 0$  has roots  $\pm \cot \frac{\pi}{10}, \pm \cot \frac{3\pi}{10}$ . **3**

(iv) Hence show that  $\cot \frac{\pi}{10} = \sqrt{5 + 2\sqrt{5}}$ . **2**

# End of paper

## Marking Scheme : Task 1 2011

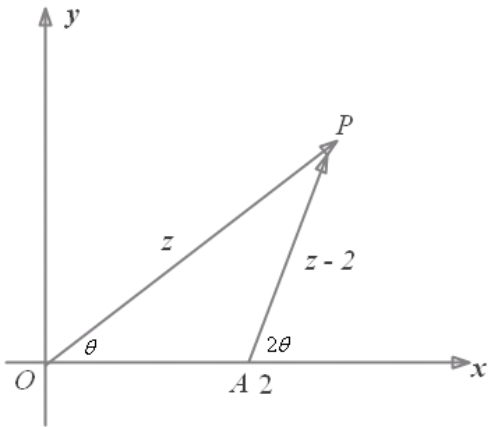
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| (a)(i) | $z_1 = 3 - 2i, z_2 = 1 + 3i.$<br>$z_1 + z_2 = 3 - 2i + 1 + 3i = 4 + i$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1 mark                                                                                                                                                                                                   | <i>Done very well</i>                                                                                                                                                                                                                                                                                          |
| (ii)   | $z_1^2 = (3 - 2i)^2$<br>$= 9 - 12i - 4$<br>$= 5 - 12i.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 mark for correct solution                                                                                                                                                                              | <i>Done very well</i>                                                                                                                                                                                                                                                                                          |
| (iii)  | $\frac{z_1}{z_2} = \frac{3 - 2i}{1 - 3i}$<br>$= \frac{3 - 2i}{1 - 3i} \times \frac{1 + 3i}{1 + 3i}$<br>$= \frac{3 + 9i - 2i + 6}{1 + 9}$<br>$= \frac{9 + 7i}{10} = \frac{9}{10} + \frac{7i}{10}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1 mark for $\overline{z_2}$<br><br><br><br><br><br><br><br><br><br>1 mark for correct answer.                                                                                                            | <i>Some students did not notice the bar on <math>z_2</math>.</i>                                                                                                                                                                                                                                               |
| b)     | $z = 1 - i.$<br>$= \sqrt{2} \operatorname{cis} \frac{-\pi}{4}$<br><br>$z^n = \sqrt{2}^n \operatorname{cis} \frac{-n\pi}{4}$<br><p>If <math>z^n</math> is purely imaginary number, then<br/> <math>\operatorname{Re}(z^n) = 0</math><br/> <math>\operatorname{Cos} \frac{-n\pi}{4} = 0</math> ie. <math>\operatorname{Cos} \frac{n\pi}{4} = 0</math><br/> <math>\therefore \frac{n\pi}{4} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots</math><br/> <math>\therefore n = 2, 6, 10, \dots</math><br/> The smallest possible value of <math>n = 2</math>.</p> <p>Or<br/> <math>\arg(z^n) = \frac{k\pi}{2}</math> where <math>k = 0, \pm 1, \pm 2, \dots</math><br/> <math>\frac{-n\pi}{4} = \frac{k\pi}{2}</math><br/> <math>n = -2k</math><br/> <math>k = -1, n = 2</math></p> | 1 mark for modulus<br><br>1 mark for argument<br><br><br><br><br><br>1 mark for setting<br>$\operatorname{Cos} \frac{-n\pi}{4} = 0$<br><br><br><br><br><br><br><br><br><br>1 mark for the correct answer | <i>The general solution of this question was incorrect some students. It was an obvious answer and therefore some students resorted to fudging the solution. All working must be shown.</i><br><br><br><br><br><br><br><br><br><br><i>Substituting <math>n = 1, 2, 3, \dots</math> is not a good strategy.</i> |

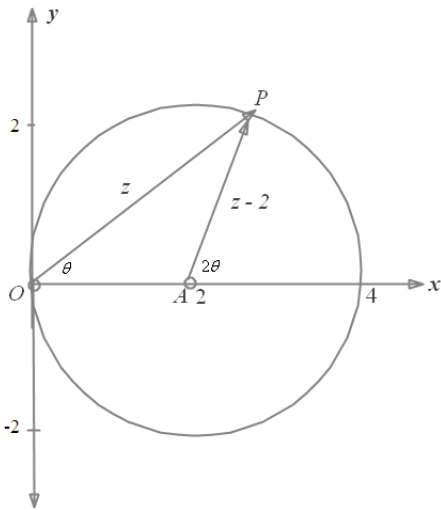
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                    |
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| c) | <p>Let <math>\sqrt{-4+3i} = x + iy</math><br/> <math>\therefore x^2 - y^2 = -4</math> (1)<br/> <math>2xy = 3 \Rightarrow xy = \frac{3}{2}</math> (2)<br/> From (1), <math>x^4 - x^2y^2 = -4x^2</math><br/> I.e. <math>x^4 - \frac{9}{4} = -4x^2</math> (using (2))<br/> <math>4x^4 + 16x^2 - 9 = 0</math><br/> <math>(2x^2 + 9)(2x^2 - 1) = 0</math><br/> <math>\therefore x^2 = -\frac{9}{2}</math> (no solution)<br/> <math>x^2 = \frac{1}{2} \therefore x = \pm \frac{1}{\sqrt{2}}</math><br/> <math>\therefore y = \pm \frac{3}{\sqrt{2}}</math><br/> <math>\sqrt{-4+3i} = \pm \frac{1}{\sqrt{2}} (1+3i)</math></p>                                                                                                                                                                                                        | <p>1 mark for equations (1) and (2).</p> <p>1 mark for the correct value of <math>x</math>.</p> <p>1 mark</p>                              | <p><i>Many students made the algebra of this too difficult due to silly errors and consequently got the incorrect answer</i></p>                                                                                                                                                                                                                                                                   |
| d) | <p><math> z+w ^2 +  z-w ^2</math><br/> <math>= (z+w) \times (\bar{z} + \bar{w}) + (z-w) \times (\bar{z} - \bar{w})</math><br/> <math>= z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w} + z\bar{z} - z\bar{w} - w\bar{z} + w\bar{w}</math><br/> <math>= 2z\bar{z} + 2w\bar{w}</math><br/> <math>= 2( z ^2) + 2( w ^2)</math><br/> <math>= 2( z ^2 +  w ^2)</math></p>  <p>As given in the diagram, <math>z + w</math> and <math>z - w</math> represent the two diagonals of the parallelogram and <math>z</math> and <math>w</math> are the lengths of the two sides. <math>OA = BC = z</math> and <math>OB = AC = w</math>.<br/> <math> z+w ^2 +  z-w ^2 = 2( z ^2 +  w ^2)</math> means <math>OC^2 + AB^2 = OA^2 + AC^2 + OB^2 + BC^2</math></p> | <p>1 mark</p> <p>1 mark</p> <p>1 mark for the diagram and explaining <math>2( z ^2 +  w ^2)</math> as the sum of squares of the sides.</p> | <p>Few students tried the algebraic method which is slightly more difficult than the method shown here.</p> <p>Diagram was shown by most students, but some had the arrows missing. Direction is essential when showing vectors.</p> <p><i>Students must make the connections <math>OA = BC = z</math> and <math>OB = AC = w</math> before writing the final conclusion. Very poorly done.</i></p> |

|  |                                                                                                                                                |                                            |  |
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|  | $\therefore$ The relationship means that in a parallelogram, the sum squares of the two diagonals is equal to the sum of squares of its sides. | 1 mark for the geometrical interpretation. |  |
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## Question 2

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| a)   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (i)  | $z_1 = 3 + i, \quad z_2 = 3 + 2\sqrt{3} + 3i$ $\overrightarrow{AB} = z_2 - z_1$ $= 3 + 2\sqrt{3} + 3i - 3 - i$ $= 2\sqrt{3} + 2i$                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1 mark                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| (ii) | $\overrightarrow{AB} = 2\sqrt{3} + 2i$ $\overrightarrow{AD} = \frac{1}{2} \overrightarrow{AB} \times \text{cis}\left(\frac{\pi}{6}\right)$ $= \frac{1}{2} (2\sqrt{3} + 2i) \times \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$ $= \frac{1}{2} (\sqrt{3} + i)^2 = \frac{1}{2} (3 + 2\sqrt{3}i - 1)$ $= 1 + \sqrt{3}i$ $\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA}$ $\therefore \overrightarrow{OD} = \overrightarrow{AD} + \overrightarrow{OA}$ $= 1 + \sqrt{3}i + 3 + i$ $= 4 + (\sqrt{3} + 1)i$ $\therefore D \text{ represents the complex number } 4 + (\sqrt{3} + 1)i$ | <p>1 mark for</p> $\overrightarrow{AD} = \frac{1}{2} \overrightarrow{AB} \times \text{cis}\left(\frac{\pi}{6}\right)$ <p>1 mark</p> <p>1 mark</p> | <p>Very poorly done by many students. When doing this question one must understand the transformation that is happening: Scale down <math>\overrightarrow{AB}</math> to <math>\frac{1}{2}</math> the length and then rotate through <math>\left(\frac{\pi}{6}\right)</math> in the anticlockwise direction (means multiply by <math>\text{cis}\left(\frac{\pi}{6}\right)</math> which <math>= \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)</math>.</p> <p>Some students misunderstood "the complex number representing D" as <math>\overrightarrow{AD}</math> rather than <math>\overrightarrow{OD}</math>.</p> |

|    |                                                                                   |                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                |
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| b) |  | <p>1 mark for the diagram<br/>         Vectors <math>OP</math> and <math>AP</math> represents <math>z</math> and <math>z - 2</math> respectively.<br/> <i>(direction of vectors must be shown)</i></p> | <p><math>z</math> means from <math>O</math> to <math>P</math> and <math>z - 2</math> means <math>A</math> to <math>P</math>. Both vectors are joined to <u><math>P</math></u>.</p> <p>Both are joined to <math>P</math>, they are not different points as some students had drawn. Directions were missing in many papers.</p> |
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| ii) | <p><math>\angle AOP = \angle APO</math>. (exterior angle equals sum of the opp. Interior angles.)</p> <p><math>\therefore OA = AP</math>.</p> <p><math>\therefore \triangle OAP</math> is isosceles.</p>  <p><math> OA  =  AP </math><br/> <math>2 =  z - 2 </math> which is a circle with centre <math>(2, 0)</math> and radius 2.</p> <p>or</p> <p>Since <math>\angle PAx = 2 \angle POx</math>, <math>P</math> must be on the circumference of the circle. (Angle subtended by arc at the centre equals double the angle at the circumference.).<br/> <math>OA = 2</math>.<br/> <math>\therefore</math> the locus of <math>P</math> is a circle with centre <math>(2, 0)</math> radius 2.</p> | <p>1 mark</p> <p>1 mark for the diagram. The points <math>O</math> and <math>A</math> must be excluded.</p> <p>1 mark for the description with reasoning.</p> | <p>Very poorly done. Many students failed to use the information <math>\arg(z) = 2\arg(z - 2)</math>.</p> <p>In general it was noted that some students did not take sufficient time to understand the question and proceeded as <math>\arg(z) - \arg(z - 2) = \theta</math>.</p> |
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| <p>c)<br/>(i)</p>  | $  \begin{aligned}  & (\cot \theta + i)^n + (\cot \theta - i)^n \\  & \left( \frac{\cos \theta}{\sin \theta} + i \right)^n + \left( \frac{\cos \theta}{\sin \theta} - i \right)^n \\  & = \frac{(\cos \theta + i \sin \theta)^n + (\cos \theta - i \sin \theta)^n}{\sin^n \theta} \\  & = \frac{(\cos n\theta + i \sin n\theta) + (\cos n\theta - i \sin n\theta)}{\sin^n \theta} \\  & = \frac{2 \cos n\theta}{\sin^n \theta}  \end{aligned}  $                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1 mark<br/>Applies DeMoivre theorem to simplify the expression</p>                         | <p>Well done in general. You must change the expression to cis<math>\theta</math> form before applying DeMoivre theorem.</p>                                                                                                                                      |
| <p>c)<br/>(ii)</p> | $  \begin{aligned}  & (x+i)^5 + (x-i)^5 \\  & \text{Let } z = \cot \theta \\  & (\cot \theta + i)^5 + (\cot \theta - i)^5 = \frac{2 \cos 5\theta}{\sin^5 \theta} = 0 \\  & ** \\  & \cos 5\theta = 0 \\  & 5\theta = 2n\pi \pm \frac{\pi}{2} = (4n \pm 1) \frac{\pi}{2}, \\  & \theta = (4n \pm 1) \frac{\pi}{10}, n = 0, \pm 1, \pm 2 \quad ** \\  & n = 0, \theta = \pm \frac{\pi}{10} \\  & z = \cot\left(\pm \frac{\pi}{10}\right) = \pm \cot\left(\frac{\pi}{10}\right) \\  & n = 1, \theta = \frac{5\pi}{10} \quad z = \cot\left(\frac{5\pi}{10}\right) = 0 \\  & \theta = \frac{3\pi}{10} \quad z = \cot\left(\frac{3\pi}{10}\right) \\  & n = -1, \theta = \frac{-3\pi}{10} \quad z = -\cot\left(\frac{3\pi}{10}\right) \\  & \therefore \text{The solutions are} \\  & 0, \pm \cot\left(\frac{\pi}{10}\right), \pm \cot\left(\frac{3\pi}{10}\right)  \end{aligned}  $ | <p>1 mark</p> <p>1 mark for the general solution.</p> <p>1 mark for the correct solutions</p> | <p>Correctly identified the substitution and applied to the result from (i).</p> <p>General solution was again a problem and students were fudging answers. In <u>show that</u> question, you must show <u>all</u> working, even the ones you can do mentally</p> |

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| (ii) | $(x+i)^5 + (x-i)^5$ $= x^5 + 5x^4i - 10x^3 - 10x^2i + 5x + i + x^5 - 5x^4i - 10x^3 + 10x^2i + 5x - i$ $= 2(x^5 - 10x^3 + 5x)$ $= 2x(x^4 - 10x^2 + 5) \quad **$ <p><math>\therefore</math> Solutions of <math>(x+i)^5 + (x-i)^5 = 0</math> are the solutions of <math>2x(x^4 - 10x^2 + 5) = 0</math>.</p> <p>Using (ii), the solutions of this polynomial are</p> $0, \pm \cot\left(\frac{\pi}{10}\right), \pm \cot\left(\frac{3\pi}{10}\right).$ $2x(x^4 - 10x^2 + 5) = 0.$ <p><math>\therefore x = 0</math> and</p> <p><math>x^4 - 10x^2 + 5 = 0</math> has</p> <p>Solutions <math>\pm \cot\left(\frac{\pi}{10}\right), \pm \cot\left(\frac{3\pi}{10}\right).</math></p> | <p>1 mark for the correct expansion and simplifies the expression to get <math>2x(x^4 - 10x^2 + 5)</math></p> <p>1 mark for <math>x = 0</math></p> <p>1 mark for solutions of <math>x^4 - 10x^2 + 5 = 0</math></p> | <p>Most students correctly did the expansion.</p>                                                                                                                           |
| (iv) | $x^4 - 10x^2 + 5 = 0$ $x^2 = \frac{10 \pm \sqrt{100 - 20}}{2}$ $= \frac{10 \pm 4\sqrt{5}}{2}$ $= 5 \pm 2\sqrt{5}$ $\therefore x = \pm \sqrt{5 \pm 2\sqrt{5}}$ <p><math>\frac{\pi}{10}</math> is in the first quadrant and</p> <p><math>\therefore \cot \frac{\pi}{10}</math> is positive.</p> <p><math>\therefore \cot \frac{\pi}{10} = \sqrt{5 + 2\sqrt{5}}</math></p>                                                                                                                                                                                                                                                                                                   | <p>1 mark for the solutions of <math>x</math></p> <p>1 mark. Reasoning required.</p>                                                                                                                               | <p>This part was done well. Students drew the graph of <math>y = \cot \theta</math> to figure out the appropriate value of <math>\cot \frac{\pi}{10}</math>. Well done.</p> |